

Acid-base, Buffers

Acid- Base Balance

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Acids and Bases Definition

- According to Bronsted, acids are substances that are capable of donating protons and bases are those that accept protons.
- Acids are proton donors and bases are proton acceptors.

Acids	Bases
$\text{HA} \leftrightarrow \text{H}^+ + \text{A}^-$	$\text{NH}_3 + \text{H}^+ \leftrightarrow \text{NH}_4^+$
$\text{HCl} \leftrightarrow \text{H}^+ + \text{Cl}^-$	$\text{HCO}_3^- + \text{H}^+ \leftrightarrow \text{H}_2\text{CO}_3$
$\text{H}_2\text{CO}_3 \leftrightarrow \text{H}^+ + \text{HCO}_3^-$	

Terms Explained

Term	Definition and explanations
pH	Negative logarithm of hydrogen ion concentration. Normal value 7.4 (range 7.38 -7.42)
Acids	Proton donors; pH <7
Bases	Proton acceptors; pH > 7
Strong acids	Acids which ionize completely; e.g. HCl
Weak acids	Acids which ionize incompletely e.g. H ₂ CO ₃
pK value	pH at which the acid is half ionized; Salt : Acid = 1:1
Alkali reserve	Bicarbonate available to neutralise acids; Normal 24 mmol/L (range 22-26 mmol/L)
Buffers	Solutions minimise changes in Ph

Weak and Strong Acids the Extent of Dissociation

- Strong acids dissociate completely in solution, while weak acids ionize incompletely
- $\text{HCl} \longrightarrow \text{H}^+ + \text{Cl}^-$ (Complete)
- $\text{H}_2\text{CO}_3 \longrightarrow \text{H}^+ + \text{HCO}_3^-$ (Partial)
- **In a solution of HCl**, almost all the molecules dissociate and exist as H^+ and Cl^- ions. Hence the concentration of H^+ is very high and it is a strong acid.
- **But in the case of a weak acid (e.g. acetic acid)**, it will ionize only partially. So, the number of acid molecules existing in the ionized state is much less, may be only 50%.

Dissociation Constant

- Dissociation of an acid is a freely reversible reaction. At equilibrium the ratio between dissociated and undissociated particle is a constant.
- Dissociation constant (K_a) of an acid is

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

where $[H^+]$ is the concentration of hydrogen ions, $[A^-]$ = the concentration of anions or conjugate base, and $[HA]$ is the concentration of undissociated molecules.

- The pH at which the acid is half ionized is called pK_a of an acid which is constant at a particular temperature and pressure.
- Strong acids will have a low pK_a and weak acids have a higher pK_a .

Acidity of a Solution and pH

- The acidity of a solution is measured by noting the hydrogen ion concentration in the solution and obtained by the equation.

$$[H^+] = K_a \frac{[\text{acid}]}{[\text{base}]} \quad \text{or} \quad \frac{[HA]}{[A^-]}$$

where K_a is the dissociation constant.

- To make it easier, Sorensen expressed the H^+ concentration as the **negative of the logarithm** (logarithm to the base 10) of hydrogen ion concentration, and is designated as the pH. Therefore,

$$pH = -\log [H^+] = \log \frac{1}{[H^+]}$$



Thus the pH value is inversely proportional to the acidity.

Lower the pH, higher the acidity or hydrogen ion concentration.

Higher the pH, the acidity is lower.

At a pH of 1, the hydrogen ion concentration is 10 times that of a solution with a pH 2 and 100 times that of a solution with a pH of 3 and so on.

The pH 7 indicates the neutral pH

Relation Between Hydrogen Ions, Hydroxyl Ions and pH of Aqueous Solutions

Ionic product of water = $[H^+][OH^-] = 10^{-14}$

$[OH^-]$ mol/liter	$[H^+]$ mol/liter	$\log$ $[H^+]$	$-\log[H^+]$ =pH	pOH	Inference
1×10^{-13}	1×10^{-1}	-1	1	13	Strong acid
1×10^{-10}	1×10^{-4}	-4	4	10	Acid
1×10^{-7}	1×10^{-7}	-7	7	7	Neutral
1×10^{-4}	1×10^{-10}	-10	10	4	Alkali
1×10^{-1}	1×10^{-13}	-13	13	1	Strong alkali

The Effect of Salt upon the Dissociation

The relationship between pH, pKa, concentration of acid and conjugate base (or salt) is expressed by the

Henderson-Hasselbalch equation,

$$\text{pH} = \text{pKa} + \log \frac{[\text{base}]}{[\text{acid}]} \quad \text{or} \quad \text{pH} = \text{pKa} + \log \frac{[\text{salt}]}{[\text{acid}]}$$

When $[\text{base}] = [\text{acid}]$; then $\text{pH} = \text{pKa}$

Therefore, when the concentrations of base and acid are the same, then pH is equal to pKa.

Thus, when the acid is half ionized, pH and pKa have the same values.

Clinical Application of Henderson-Hasselbalch's Equation

- The pH of a buffer on addition of a known quantity of acid and alkali can, therefore, be predicted by the equation.
- Moreover, the concentration of salt or acid can be found out by measuring the pH.
- The Henderson-Hasselbalch's equation, therefore, has great practical application in clinical practice in assessing the acid-base status, and predicting the limits of the compensation of body buffers.

Buffers

- Buffers are solutions which can resist changes in pH when acid or alkali is added.
- Buffers are of two types:
 - a. Mixtures of weak acids with their salt with a strong base or
 - b. Mixtures of weak bases with their salt with a strong acid.
- i. $\text{H}_2\text{CO}_3 / \text{NaHCO}_3$ (Bicarbonate buffer)
(carbonic acid and sodium bicarbonate)
- ii. $\text{CH}_3\text{COOH} / \text{CH}_3\text{COO Na}$ (Acetate buffer)
(acetic acid and sodium acetate)
- iii. $\text{Na}_2\text{HPO}_4 / \text{NaH}_2\text{PO}_4$ (Phosphate buffer)

Factors Affecting pH of a Buffer

a) The value of pK :

The lower the value of pK , the lower is the pH of the solution.

a) The ratio of salt to acid concentrations

Actual concentrations of salt and acid in a buffer solution may be varying widely, with no change in pH, so long as the ratio of the concentrations remains the same.

Factors Affecting Buffer Capacity

- The buffering capacity of a buffer is defined as the ability of the buffer to resist changes in pH when an acid or base is added.
- Buffer capacity is determined by the actual concentrations of salt and acid present, as well as by their ratio.
- Buffering capacity is the number of grams of strong acid or alkali which is necessary for a change in pH of one unit of one liter of buffer solution.

How do Buffers Act?

- Buffer solutions consist of mixtures of a weak acid or base and its salt.
- When hydrochloric acid is added to the acetate buffer, the salt reacts with the acid forming the weak acid, acetic acid and its salt.



Thus changes in the pH are minimized.

- Similarly when a base is added, the acid reacts with it forming salt and water.



The Buffer Capacity is Determined by the Absolute Concentration of the Salt and Acid

- But the pH of the buffer is dependent on the relative proportion of the salt and acid (Henderson-Hasselbalch's equation).
- The pH of a buffer on addition of a known quantity of acid and alkali can, therefore, be predicted by the equation.
- The Henderson-Hasselbalch's equation therefore, has great practical application in clinical practice in assessing the acid-base status, and predicting the limits of the compensation of body buffers.

Effective Range of a Buffer

- A buffer is most effective when the concentrations of salt and acid are equal or when $\text{pH} = \text{pK}_a$.
- The effective range of a buffer is 1 pH unit higher or lower than pK_a .
- Since the pK_a values of most of the acids produced in the body are well below the physiological pH, they immediately ionize and add H^+ to the medium.
- This would necessitate effective buffering.

Acid-base Balance

Normal pH of plasma is 7.4.

In normal life, the variation of plasma pH is very small.

The pH of plasma is maintained within a narrow range of 7.38 to 7.42.

The pH of the interstitial fluid is generally 0.5 units below that of the plasma.



Acidosis

- If the pH is below 7.38, it is called **acidosis**.
- Life is threatened when the pH is lowered below 7.25.
- Acidosis leads to **CNS depression and coma**.
- Death occurs when pH is below 7.0.

Alkalosis

- When the pH is more than 7.42, it is **alkalosis**.
- It is very dangerous if pH is increased above 7.55.
- Alkalosis induces neuromuscular **hyperexcitability** and tetany.
- Death occurs when the pH is above 7.6.

➤ Normal pH

- The **pH of plasma is 7.4**.
- The pH of plasma is maintained within a **narrow range of 7.38 to 7.42**; but for clinical purposes, the range used is 7.35 to 7.45. The pH of the interstitial fluid is generally 0.5 units below that of the plasma.
- The intracellular pH is lower than the extracellular pH.

➤ Acidemia or Acidosis

- Acidosis is a pathological state caused by a disease having abnormal increase in acid in the body.
- Whereas acidemia is defined as decreased blood pH.
- If the pH is below 7.35, it is called acidemia.
- Life is threatened when the pH is lowered below 7.25.
- Death occurs when pH is below 7.0.



➤ **Alkalemia or Alkalosis**

- Alkalosis is a pathological state caused by a disease having abnormal decrease in acid in the body.
- Whereas alkalemia is defined as increased blood pH.
- When the pH is more than 7.45, it is called alkalemia.
- It is very dangerous if pH is increased above 7.55.
- Alkalosis induces neuromuscular hyperexcitability and tetany.
- Death occurs when the pH is above 7.6.

Mechanisms of Regulation of pH

- First line of defense: Blood buffers
- Second line : Respiratory regulation
- Third line of defense: Renal regulation

Buffers of the Body Fluids

Buffers are the **first line of defense** against acid load.

The buffers are effective as long as the acid load is not excessive, and the alkali reserve is not exhausted.

A) Bicarbonate Buffer System

The most important buffer system in the plasma is the bicarbonate-carbonic acid system ($\text{NaHCO}_3 / \text{H}_2\text{CO}_3$).

It accounts for 65% of buffering capacity in plasma and 40% of buffering action in the whole body.



The base constituent, bicarbonate(HCO_3^-), is regulated by the kidney (metabolic component).

Acid part, carbonic acid (H_2CO_3), is under respiratory regulation (respiratory component).

The normal bicarbonate level of plasma is 24 mmol/liter.

The normal pCO_2 of arterial blood is 40 mm of Hg.

The pKa for carbonic acid is 6.1. Substituting these values in the Henderson–Hasselbalch's equation,

$$\text{pH} = \text{pKa} + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$= 6.1 + \log \frac{24}{1.2}$$

$$= 6.1 + \log 20$$

$$= 6.1 + 1.3$$

$$= 7.4$$



Hence, the ratio of HCO_3^- to H_2CO_3 at pH 7.4 is 20 under normal conditions. This is much higher than the theoretical value of 1 which ensures maximum effectiveness.

The bicarbonate carbonic acid buffer system is the most important for the following reasons:

- a. Presence of bicarbonate in relatively high concentrations.
- b. The components are under physiological control, CO_2 by lungs and bicarbonate by kidneys

Alkali Reserve

Bicarbonate represents the alkali reserve It has to be sufficiently high to meet the acid load.

If it was too low to give a ratio of 1, all the HCO_3^- would have been exhausted within a very short time; and buffering will not be effective.

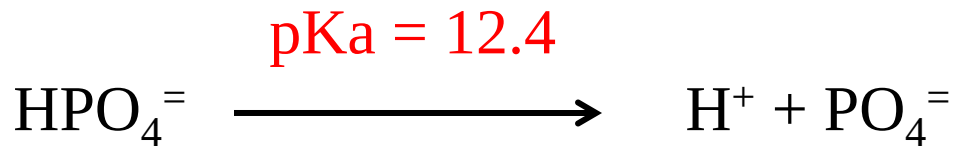
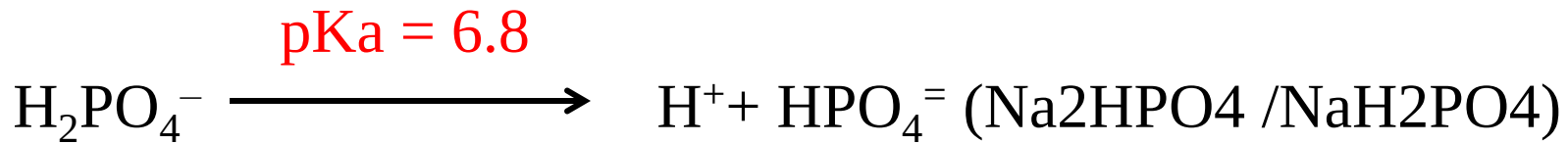
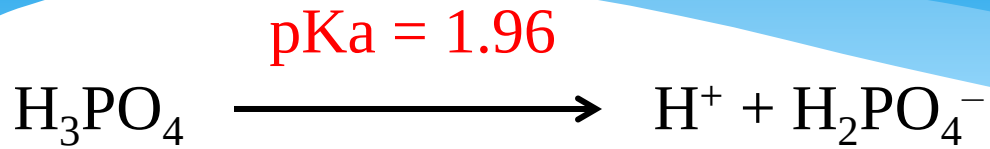
So, under physiological circumstances, the ratio of 20 (a high alkali reserve) ensures high buffering efficiency against acids.

B) Phosphate Buffer System

It is mainly an **intracellular** buffer. Its concentration in plasma is very low.

The pKa value is 6.8.

The phosphate buffer system is found to be effective at a wide pH range, because it has more than one ionizable group and the pKa values are different for both.



In the body, Na_2HPO_4 / NaH_2PO_4 is an effective buffer system, because its pKa value is nearest to physiological pH.

C) Protein Buffer System

Buffering capacity of protein depends on the pKa value of ionizable side chains.

The most effective group is histidine imidazole group with a pKa value of 6.1.

Hemoglobin buffer

Buffer Systems of the Body

Extracellular fluid	Intracellular fluid	Erythrocyte fluid
<u>NaHCO_3</u> H_2CO_3 (bicarbonate)	<u>K_2HPO_4</u> KH_2PO_4 (phosphate)	<u>$\text{K}+\text{Hb}$</u> $\text{H}+\text{Hb}$ (hemoglobin)
<u>Na_2HPO_4</u> NaH_2PO_4 (phosphate)	<u>$\text{K}+\text{Protein}$</u> $\text{H}^+\text{Protein}$ (protein buffer)	<u>K_2HPO_4</u> KH_2PO_4 (phosphate)
<u>$\text{Na} + \text{Albumin}$</u> $\text{H}^+\text{Albumin}$	<u>KHCO_3</u> H_2CO_3	<u>KHCO_3</u> H_2CO_3

Cellular Buffers

Cytoplasmic pH varies from 6.8 to 7.3.

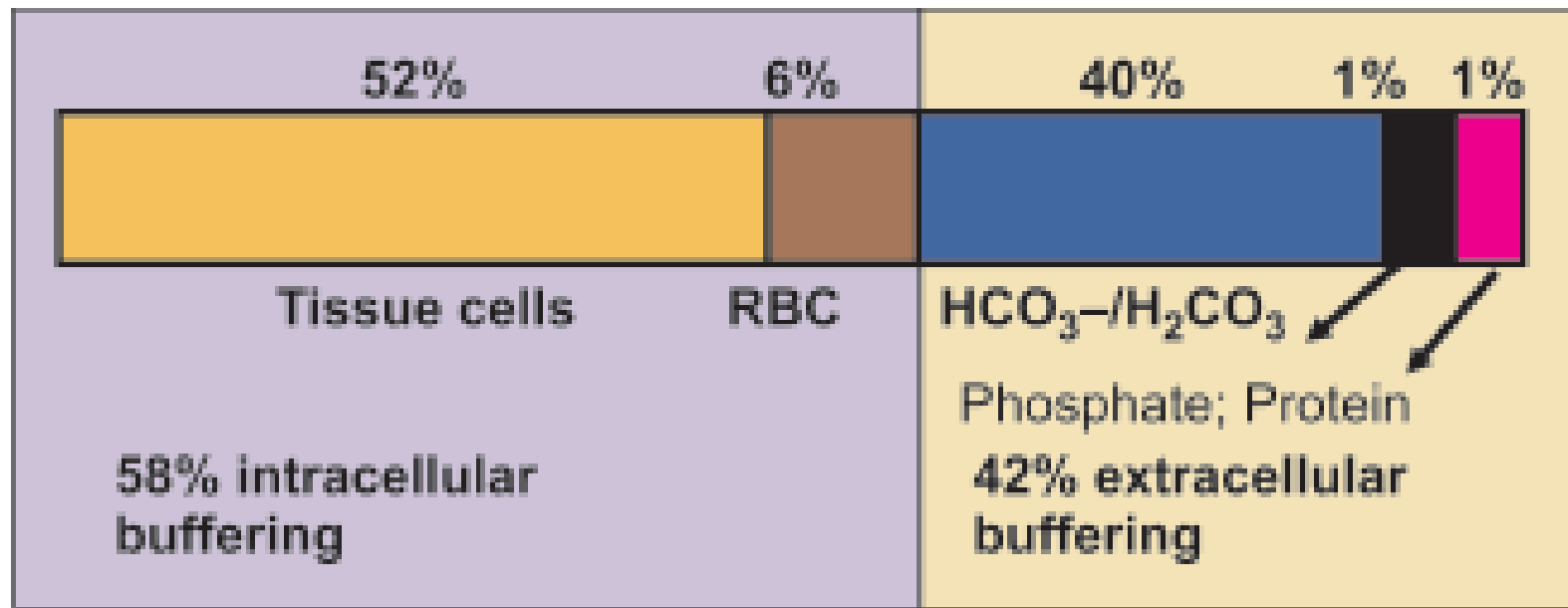
Intracellular pH modulates cell functions:

1. Several enzymes is sensitive to changes in pH.
2. Reduction in pH reduces contractility of actin and myosin in muscles.
3. The electrical properties of excitable cells are also affected by changes in pH. Major tissues involved in cellular buffering are bone and skeletal muscle.

Relative Capacity of Buffer Systems

In the body, 52% buffer activity is in tissue cells and 6% in RBCs; 43% is by extracellular buffers.

In plasma and extracellular space, about 40% buffering action is by bicarbonate system; 1% by proteins and 1% by phosphate buffer system.



Buffers Act Quickly, But Not Permanently

Buffers can respond immediately to addition of acid or base, but they do not serve to eliminate the acid from the body.

They are also unable to replenish the alkali reserve of the body.

For the final elimination of acids, the respiratory and renal regulations are very essential.

2. Respiratory Regulation of pH

The Second Line of Defense

By changing the $p\text{CO}_2$ (or carbonic acid, the denominator in the equation).

The CO_2 diffuses from the cells into the extracellular fluid and reaches the lungs through the blood.

The rate of respiration (rate of elimination of CO_2) is controlled by chemoreceptors in the respiratory center which are sensitive to changes in the pH of blood.

2. Respiratory Regulation of pH

The Second Line of Defense

When there is a fall in pH of plasma (acidosis), the respiratory rate is stimulated resulting in hyperventilation.

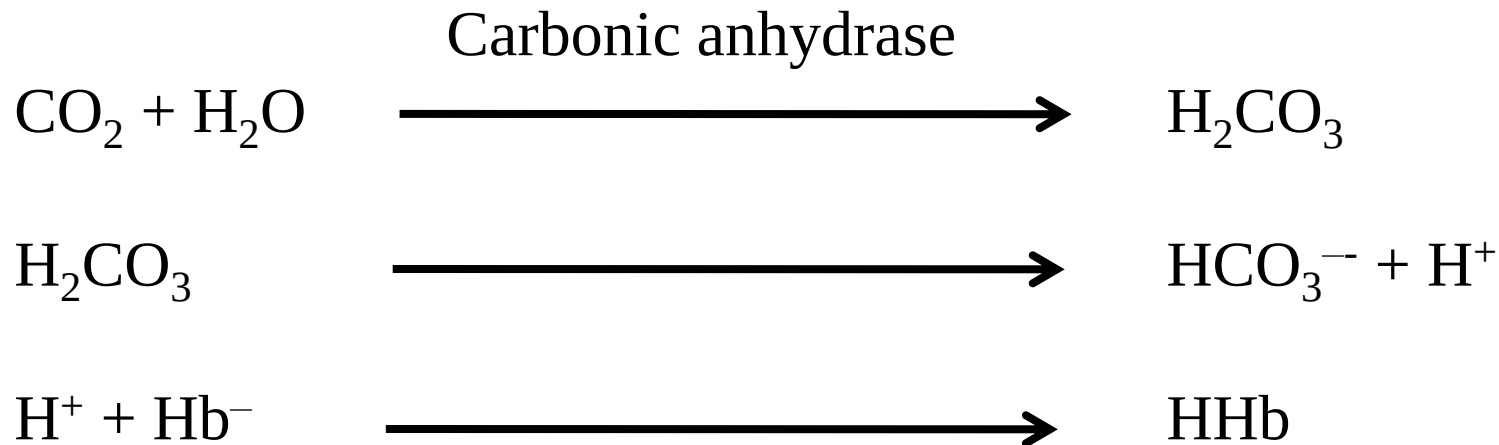
This would eliminate more CO_2 , thus lowering the H_2CO_3 level.

However, this cannot continue for long. The respiratory system responds to any change in pH immediately, but it cannot proceed to completion.

Action of Hemoglobin

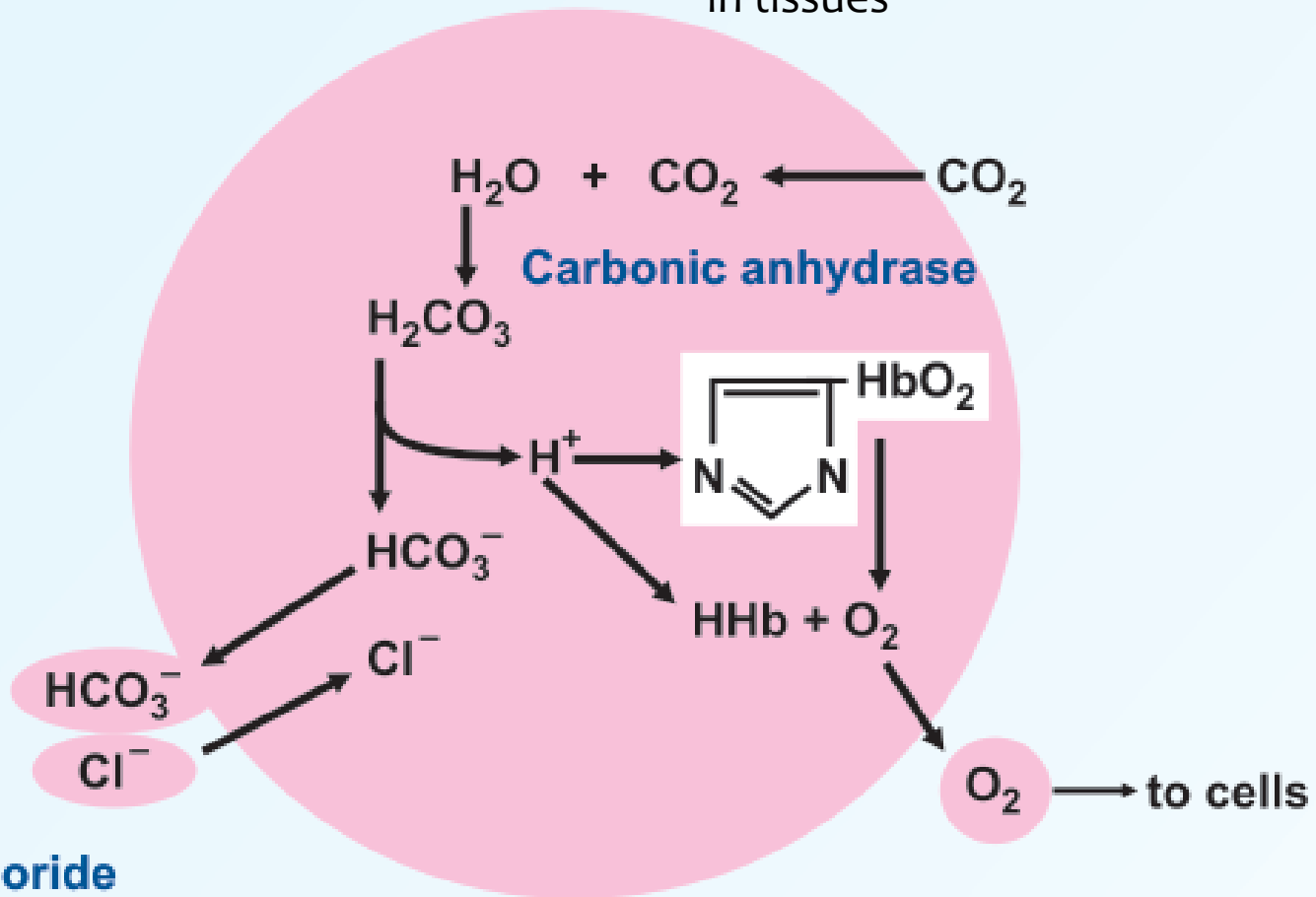
Hemoglobin serves to transport the CO_2 formed in the tissues, with minimum change in pH. (isohydric transport).

Side by side, it serves to generate bicarbonate or alkali reserve by the activity of the carbonic anhydrase system.



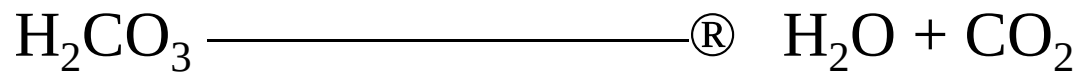
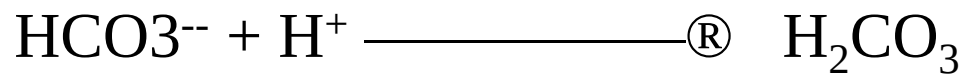
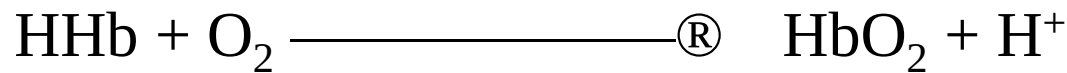
Erythrocyte in tissue capillary

Chloride shift; reactions in tissues



Chloride enters into RBC

The reverse occurs in the lungs during oxygenation and elimination of CO₂. When blood reaches the lungs, the Bicarbonate re-enters the erythrocytes.

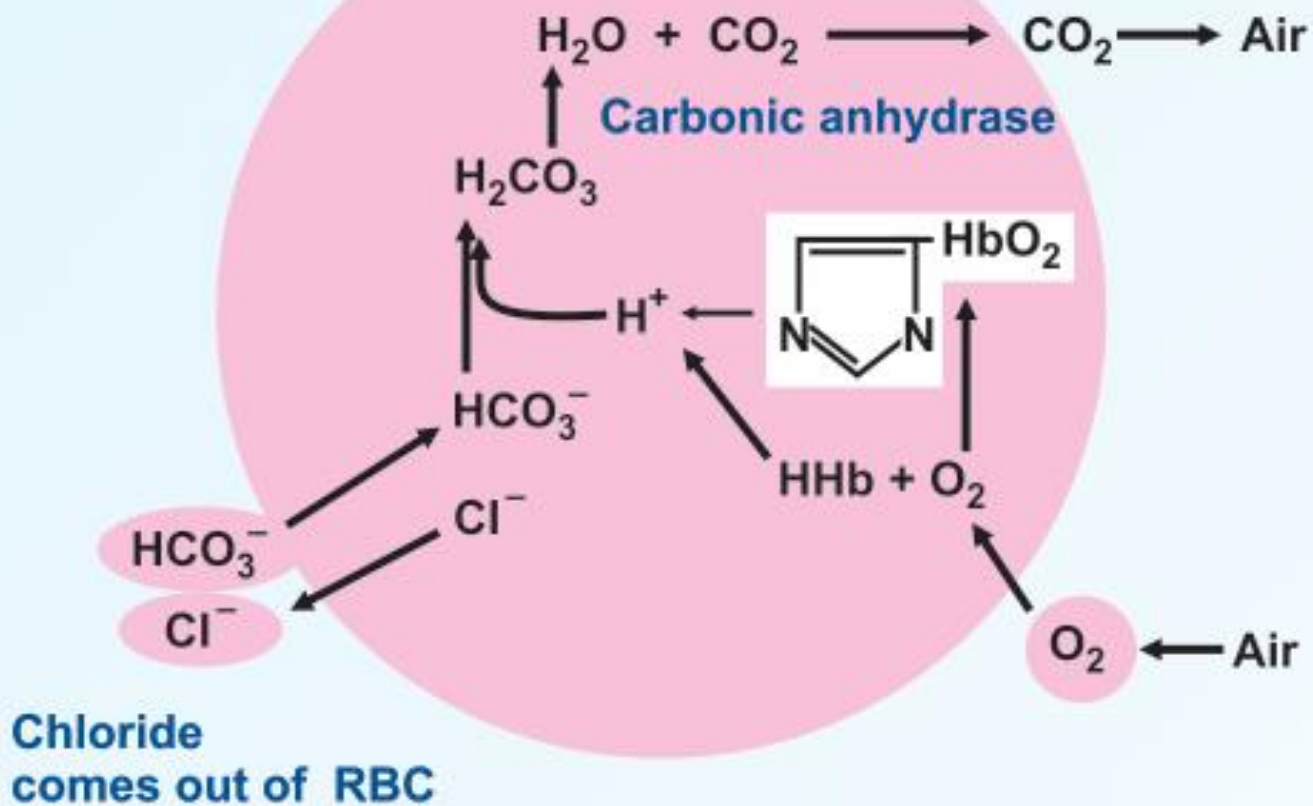


CO₂ is thus eliminated by the lungs.

The activity of the carbonic anhydrase (carbonate dehydratase) increases in acidosis and decreases with decrease in H⁺ concentration.

Erythrocyte in lung capillary

Chloride shift;
reactions in lungs



3. Renal Regulation of pH

Normal urine has a pH around 6; this pH is lower than that of extracellular fluid (pH = 7.4).

This is called **acidification of urine**.

The pH of the urine may vary from as low as 4.5 to as high as 9.8, depending on the amount of acid excreted.

Major Renal Mechanisms:

A. Excretion of H^+

B. Reabsorption of bicarbonate (recovery of bicarbonate)

C. Excretion of titratable acid (net acid excretion)

D. Excretion of NH_4^+ (ammonium ions) .

3.A. Excretion of H^+ ; Generation of Bicarb

Occurs in proximal convoluted tubules.

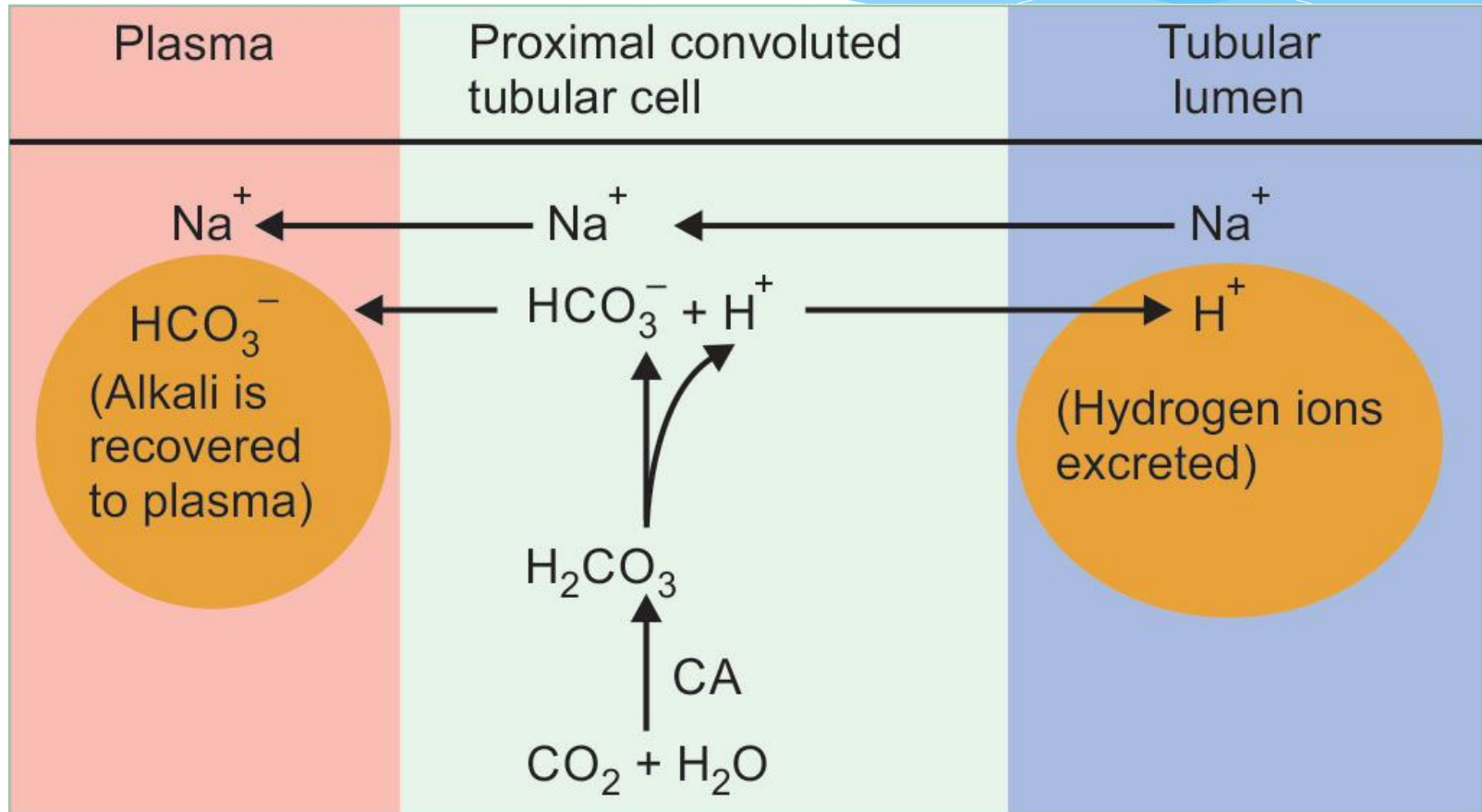
CO_2 combines with water to form carbonic acid, with the help of carbonic anhydrase. H_2CO_3 then ionizes to H^+ and bicarbonate.

The hydrogen ions are secreted into the tubular lumen; in exchange for Na^+ reabsorbed.

These Na^+ ions along with HCO_3^- will be reabsorbed into the blood.

There is net excretion of hydrogen ions, and net generation of bicarbonate.

So this mechanism serves to increase the alkali reserve.



Excretion of Hydrogen Ions in the Proximal Tubules

3.B. Reabsorption of Bicarbonate

This is mainly a mechanism to conserve base. There is no net excretion of H^+

When Na^+ enters the cell, hydrogen ions from the cell are secreted into the luminal fluid. The hydrogen ions are generated within the cell by the action of carbonic anhydrase.

The hydrogen ions secreted into the luminal fluid is required for reabsorption of filtered bicarbonate.

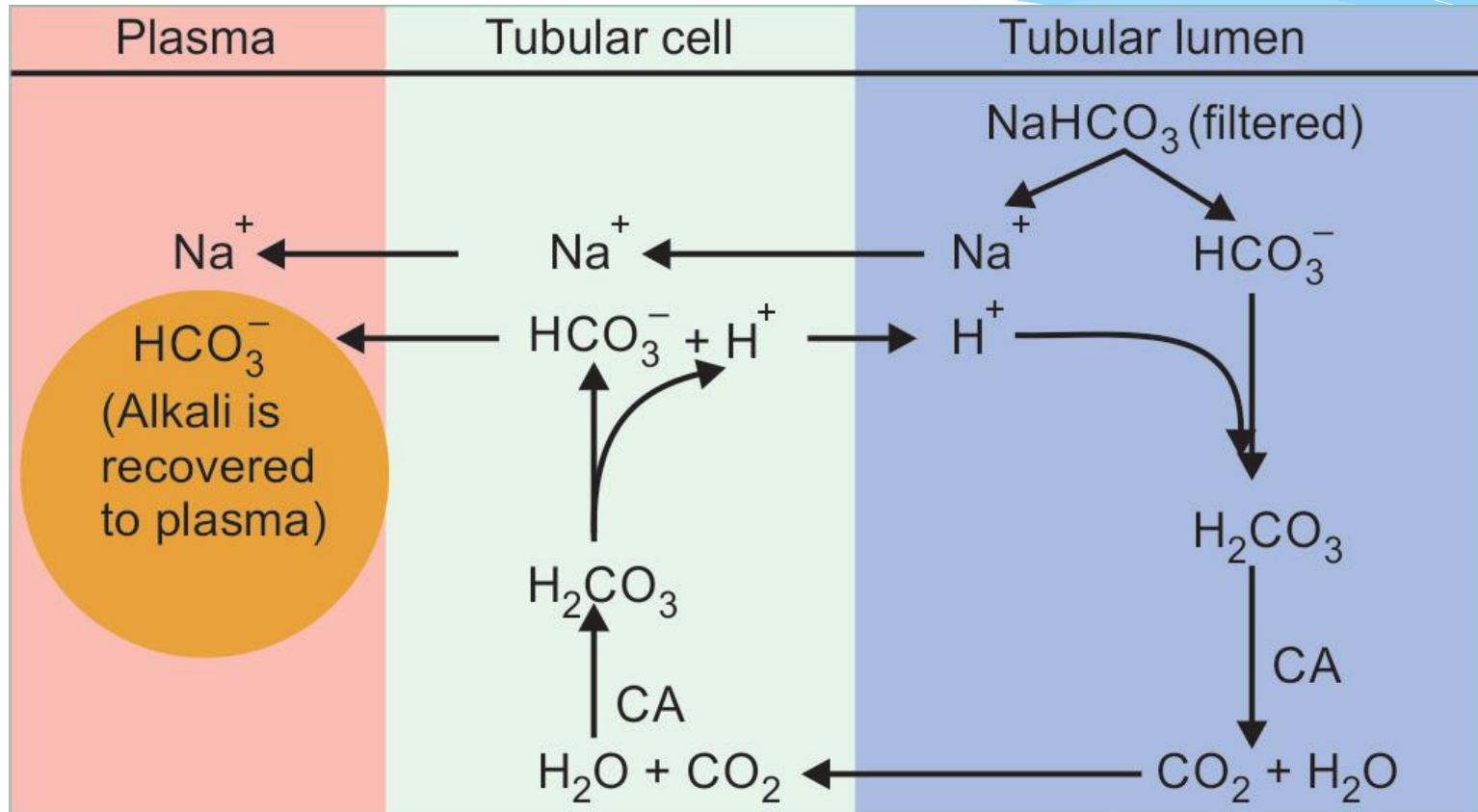
Bicarbonate is filtered by the glomerulus. This is completely reabsorbed by the proximal convoluted tubule, so that the urine is normally bicarbonate free.

Bicarbonate combines with H^+ in tubular fluid to form carbonic acid. It dissociates into water and CO_2 . The CO_2 diffuses into the cell, which again combines with water to form carbonic acid.

In the cell, it again ionizes to H^+ that is secreted into lumen in exchange for Na^+ . The HCO_3^- is reabsorbed into plasma along with Na^+ .

Here, there is no net excretion of H^+ or generation of new bicarbonate.

Net effect is the reabsorption of filtered bicarbonate. This mechanism prevents the loss of bicarbonate through urine.



Reabsorption of Bicarbonate from the Tubular Fluid

3.C. Excretion of H^+ as Titratable Acid

In the distal convoluted tubules net acid excretion occurs.

Hydrogen ions are generated in the tubular cell by carbonic anhydrase.

The bicarbonate generated within the cell passes into plasma.

The term titratable acidity of urine refers to the number of milliliters of N/10 NaOH required to titrate 1 liter of urine to pH 7.4.

This is a measure of net acid excretion by the kidney.

3.C. Excretion of H^+ as Titratable Acid

The major titratable acid present in the urine is sodium acid phosphate.

As the tubular fluid passes down the renal tubules more and more H^+ are secreted into the luminal fluid so that its pH steadily falls.

The process starts in the proximal tubules, but continues upto the distal tubules.

3.C. Excretion of H^+ as Titratable Acid

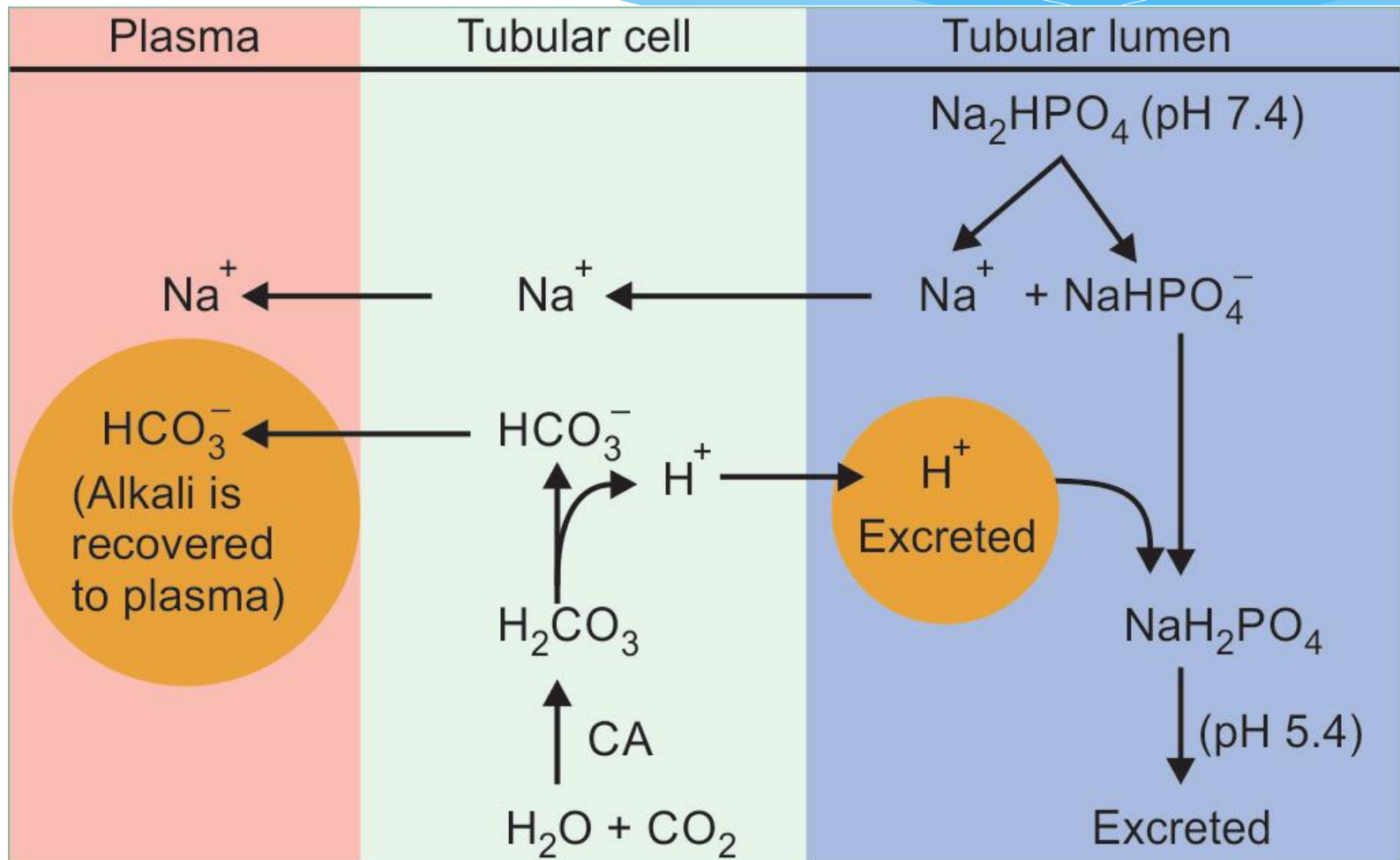
Due to the Na^+ to H^+ exchange occurring at the renal tubular cell border, the Na_2HPO_4 (basic phosphate) is converted to NaH_2PO_4 (acid phosphate) .

As a result, the pH of tubular fluid falls.

The acid and basic phosphate pair is considered as the urinary buffer.

The maximum limit of acidification is pH 4.5.

This process is inhibited by carbonic anhydrase inhibitors like acetazolamide.



Phosphate Mechanism in Tubules

3.D. Excretion of Ammonium Ions

Predominantly occurs at the distal convoluted tubules.

This would help to excrete H^+ and reabsorb HCO_3^-

This mechanism also helps to trap hydrogen ions in the urine, so that large quantity of acid may be excreted with minor changes in pH.

The excretion of ammonia helps in the elimination of hydrogen ions without appreciable change in the pH of the urine.

3.D. Excretion of Ammonium Ions

The Glutaminase present in the tubular cells can hydrolyze glutamine to ammonia and glutamic acid.

The NH_3 (ammonia) diffuses into the luminal fluid and combines with H^+ to form NH_4^+ (ammonium ion).

Glutaminase activity is increased in acidosis.

So large quantity of H^+ ions are excreted as NH_4^+ in acidosis.

3.D. Excretion of Ammonium Ions

Since it is a positively charged ion, it can accompany negatively charged acid anions; so Na^+ and K^+ are conserved.

Normally, about 70 mEq/L of acid is excreted daily; but in condition of acidosis, this can rise to 400 mEq/day.

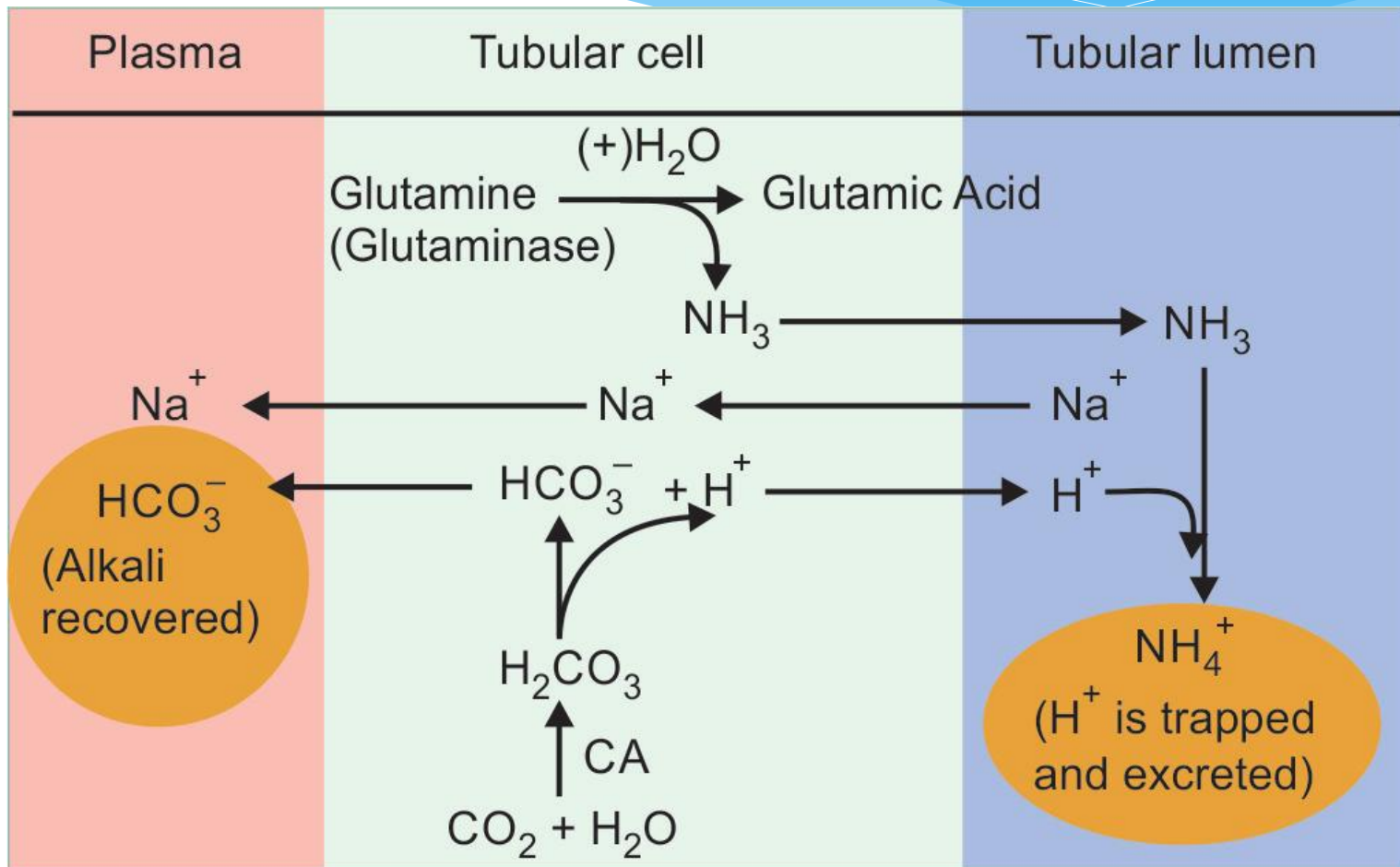
The enhanced activity of glutaminase and increased excretion of NH_4^+ takes about 3-4 days to set in under conditions of acidosis. But once established, it has high capacity to eliminate acid.

3.D. Excretion of Ammonium Ions

Ammonia is estimated in urine, after addition of formaldehyde.

The titratable acidity plus the ammonia content will be a measure of acid excreted from the body.

Maximum urine acidity reached is 4.4.



Ammonia Mechanism

Factors affecting Renal Acid Excretion

1. Increased filtered load of bicarbonate
2. Decrease in ECF volume
3. Decrease in plasma pH
4. Increase in $p\text{CO}_2$ of blood
5. Hypokalemia
6. Aldosterone secretion

Relationship of pH with K⁺ Ion Balance

Buffering of acid is achieved by the exchange of H⁺ that enters into the cells for Na⁺ or K⁺ ions.

When there is increase in H⁺ in extracellular fluid (ECF), there may be exchange of H⁺ with K⁺ from within the cells.

Net effect is an apparent increase in ECF potassium level (hyperkalemia).

In general, metabolic acidosis is associated with hyperkalemia and metabolic alkalosis with hypokalemia.

Relationship of pH with K⁺ Ion Balance

However, in renal tubular acidosis, due to failure to excrete hydrogen ions, potassium is lost in urine; then hypokalemia results.

Sudden hypokalemia may develop during the correction of acidosis.

K⁺ may go back into the cells, suddenly lowering the plasma K⁺.

Hence, it is important to maintain the K⁺ balance during correction of acidosis.

Buffering against Acid Load

Stages	Features	Buffer components
Normal	Normal ratio = 20:1 Normal pH = 7.4	$\frac{\text{HCO}_3^- \text{ (N)}}{\text{H}_2\text{CO}_3 \text{ (N)}}$
First line of defense Plasma buffer system	Acidosis; H^+ enters blood, bicarbonate is used up	$\text{HCO}_3^- (\downarrow\downarrow)$
Second line defense Respiratory compensation	Hyperventilation $\text{H}_2\text{CO}_3 \rightarrow \text{H}_2\text{O} + \text{CO}_2 \uparrow$	$\text{H}_2\text{CO}_3 (\downarrow)$
Partially compensated acidosis	Bicarbonate $\downarrow$; pH $\downarrow$	$\frac{\text{HCO}_3^- (\downarrow\downarrow)}{\text{H}_2\text{CO}_3 (\downarrow\downarrow)}$
Third line of defense kidney mechanism	Excretion of H^+ ; Reabsorption of bicarbonate; Ratio and pH tend to restore	$\frac{\text{HCO}_3^- (\downarrow\downarrow)}{\text{H}_2\text{CO}_3 (\downarrow\downarrow)}$

Acid-base Parameters are to be Checked in Patients with

1. Any serious illness
2. Multi-organ failure
3. Respiratory failure
4. Cardiac failure
5. Uncontrolled diabetes mellitus
6. Poisoning (barbiturates, ethylene glycol)

Steps to the Clinical Assessment

1. Assess pH (normal 7.4); pH <7.38 is acidemia and >7.42 is alkalemia
2. Serum bicarbonate level.
3. Assess arterial pCO₂.
4. Check compensatory response:
Compensation never overcorrects the pH.
If pH is <7.4 , acidosis is the primary disorder.
If pH is >7.4 , alkalosis is primary.
5. Assess anion gap.
6. Assess the change in serum anion gap/change in bicarbonate.
7. Assess if there is any underlying cause.

Classification of Acid-Base Disturbances

1. Acidosis (fall in pH)

- a. **Respiratory acidosis**: Primary excess of carbonic acid.
- b. **Metabolic acidosis**: Primary deficit of bicarbonate.

2. Alkalosis (Rise in pH)

- a. **Respiratory alkalosis**: Primary deficit of carbonic acid.
- b. **Metabolic alkalosis**: Primary excess of bicarbonate

3. Compensatory responses

- a. Uncompensated
- b. Partially compensated
- c. Fully compensated.

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b. Partially compensated

c. Fully compensated.

Compensatory Responses

Primary disturbances will be followed by a secondary compensatory change in the counteracting variable, e.g.

a primary change in bicarbonate involves an alteration in $p\text{CO}_2$.

Patients will have different states of compensation

The compensatory (adaptive) responses are:

Renal and Respiratory Compensations

Metabolic acidosis: Expect $p\text{CO}_2$ to be reduced by 1 mmHg for every 1 mmol/L drop in bicarbonate.

Metabolic alkalosis: Expect $p\text{CO}_2$ to be increased by 0.6 mmHg for every 1 mmol/L rise in bicarbonate.

Acute respiratory acidosis: Expect 1 mmol/L increase in bicarbonate per 10 mmHg rise in $p\text{CO}_2$.

Renal and Respiratory Compensations

Chronic respiratory acidosis: Expect 3.5 mmol/L increase in bicarbonate per 10 mmHg rise in pCO₂.

Acute respiratory alkalosis: Expect 2 mmol/L decrease in bicarbonate per 10 mmHg fall in pCO₂.

Chronic respiratory alkalosis: Expect 4 mmol/L decrease in bicarbonate per 10 mmHg fall in pCO₂.

Types of Acid-base Disturbances

Distur-bance	pH	Primary change	Ratio	Secondary change
Metabolic Acidosis	Decrease	Deficit of bicarb	<20	Decrease in PaCO₂
Metabolic alkalosis	Increased	Excess of bicarb	>20	Increase in PaCO₂
Respiratory acidosis	Decrease	Excess of carbonic acid	<20	Increase in bicarb
Respiratory alkalosis	Increased	Deficit of carbonic acid	>20	Decrease in bicarb

Clinically, Acid-base Disturbance States may be Divided into:

- i. Uncompensated
- ii. Partially compensated
- iii. Fully compensated.

Primary decrease in bicarbonate involves a reduction in blood $p\text{CO}_2$ by alveolar hyperventilation.

Primary increase in arterial $p\text{CO}_2$ involves an increase in bicarbonate by an increase in bicarbonate reabsorption by the kidney.

The compensatory change will try to restore the pH to normal.

However, the compensatory change cannot fully correct a disturbance.

Mixed Responses

- i. In mixed disturbances, both HCO_3^- and H_2CO_3 levels are altered
- ii. The adaptive response always involves a change in the counteracting variable; e.g. a primary change in bicarbonate involves an alteration in pCO_2 .
- iii. Depending on the extent of the compensatory change there are different stages.

Acid-base Disturbances: Expected Renal and Respiratory Compensations

- **Metabolic Acidosis:** Expect $p\text{CO}_2$ to be reduced by 1 mm Hg for every 1 mmol/L drop in bicarbonate.
- **Metabolic Alkalosis:** Expect $p\text{CO}_2$ to be increased by 0.6 mm Hg for every 1 mmol/L rise in bicarbonate.
- **Acute Respiratory Acidosis:** Expect 1 mmol/L increase in bicarbonate per 10 mm Hg rise in $p\text{CO}_2$.
- **Chronic Respiratory Acidosis:** Expect 3.5 mmol/L increase in bicarbonate per 10 mm Hg rise in $p\text{CO}_2$.
- **Acute Respiratory Alkalosis:** Expect 2 mmol/L decrease in bicarbonate per 10 mm Hg fall in $p\text{CO}_2$.
- **Chronic Respiratory Alkalosis:** Expect 4 mmol/L decrease in bicarbonate per 10 mm Hg fall in $p\text{CO}_2$.

Stages of Compensation

Stage	pH	HCO ₃	PaCO ₂	Ratio
Metabolic acidosis	Low	Low	N	< 20
Partially compensated	Low	Low	Low	< 20
Fully compensated	N	Low	Low	20
Metabolic alkalosis	High	High	N	> 20
Partially compensated	High	High	High	> 20
Fully compensated	N	High	High	20

Stage	pH	HCO ₃	PaCO ₂	Ratio
Respiratory acidosis	Low	N	High	<20
Partially compensated	Low	High	High	<20
Fully compensated	N	High	High	20
Respiratory alkalosis	High	N	Low	>20
Partially compensated	High	Low	Low	>20
Fully compensated	N	Low	Low	20

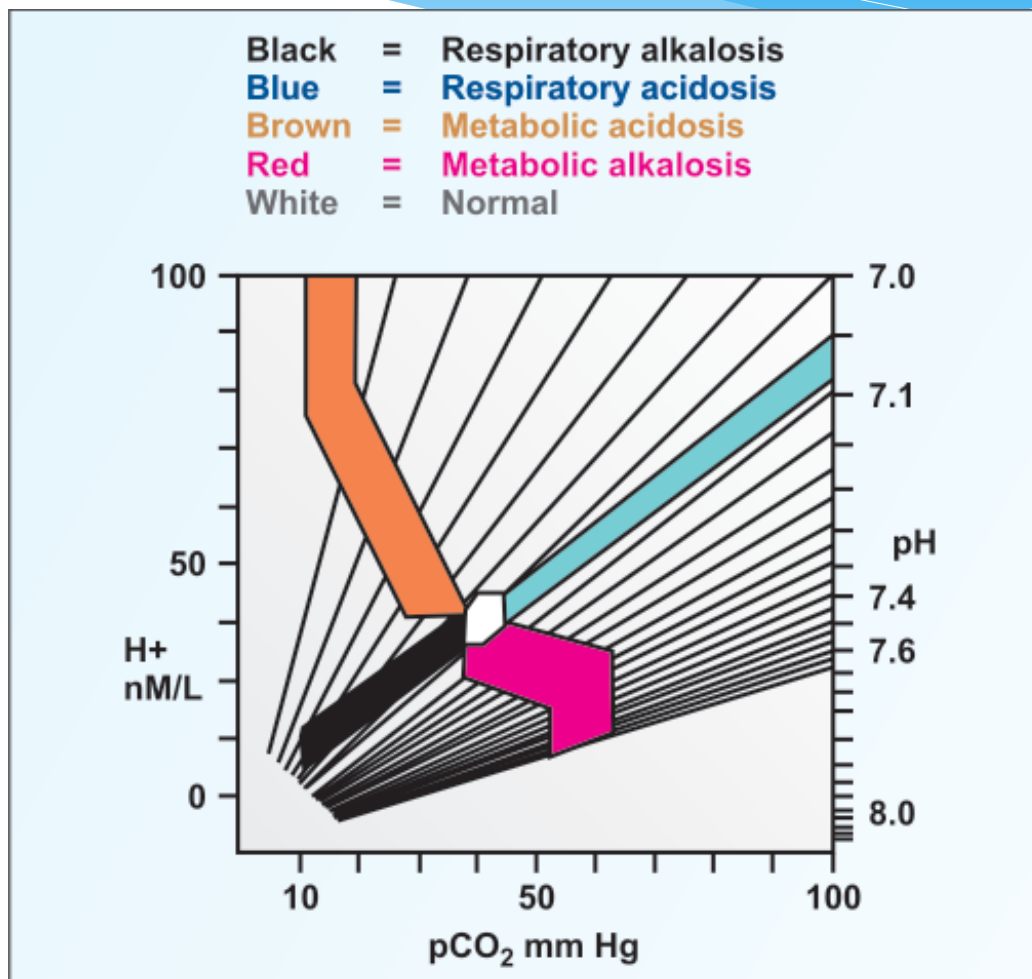


Fig. 29.6. Bicarbonate Diagram

Anion Gap

The sum of cations and anions in ECF is always equal, so as to maintain the electrical neutrality.

Sodium and potassium together account for 95% of the cations whereas chloride and bicarbonate account for only 86% of the anions.

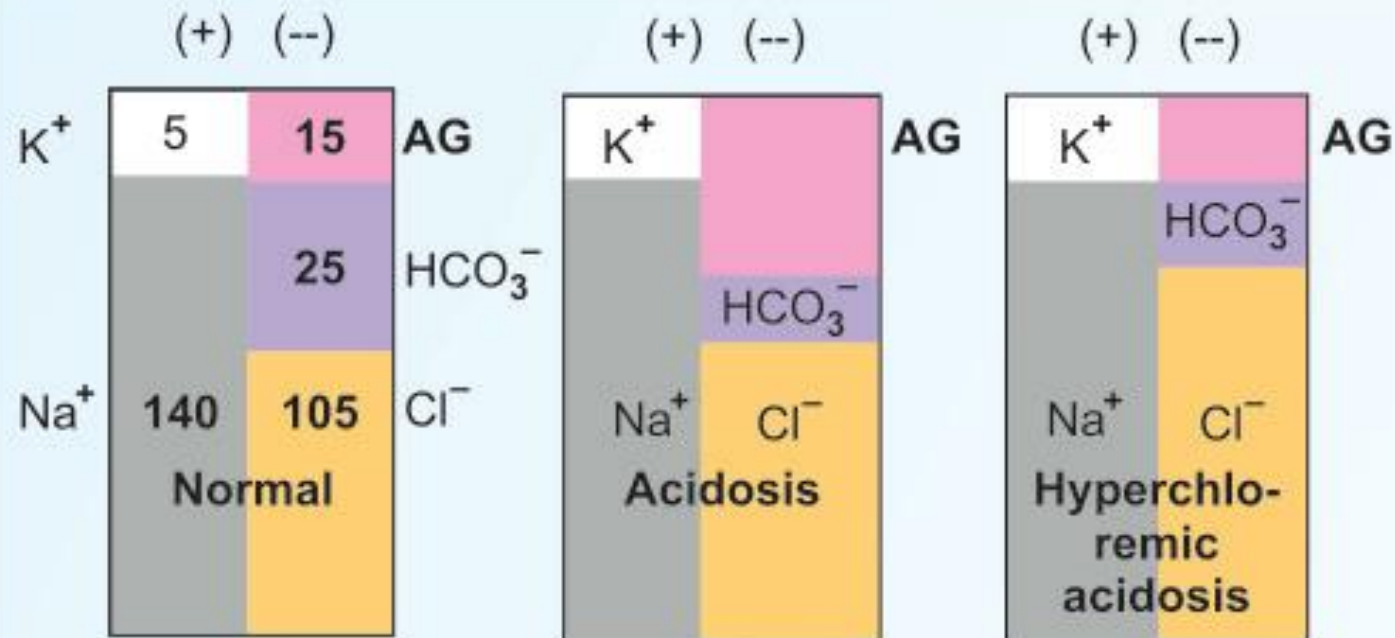
Only these electrolytes are commonly measured.

Hence there is always a difference between the measured cations and the anions. **The unmeasured anions constitute the anion gap.**

This is due to the presence of protein anions, sulphate, phosphate and organic acids.

The anion gap is calculated as $(\text{Na}^+ + \text{K}^+)$ minus $(\text{HCO}_3^- + \text{Cl}^-)$.

Normally this is about 12 mmol/liter.



The figures are the concentration in mEq/l or mmol/L. The red shaded area denotes **AG (anion gap)**. In acidosis, the bicarbonate is reduced causing an increase in the anion gap. In hyperchloremic acidosis, there is no change in the anion gap, but as a compensation, chloride ions are increased.

High Anion Gap Metabolic Acidosis (HAGMA)

Accumulation of acid anions or, cations are decreased (hypokalemia, hypo-calcemia, hypomagnesemia).

Hypoalbuminemia (decrease in negatively charged protein),
hypergammaglobulinemia (increase in positively charged protein).

Renal Failure

The excretion of H^+ as well as generation of bicarbonate are both deficient. The anion gap increases due to accumulation of other buffer anions.

Diabetic ketoacidosis.

Lactic acidosis: Normal lactic acid content in plasma is less than 2 mmol/L. It is increased in tissue hypoxia, circulatory failure, and intake of biguanides.

Lactic acidosis causes a raised anion gap, whereas diarrhea causes a normal anion gap acidosis

Suppose **lactic acid** has entered in blood; this is buffered by bicarbonate, resulting in sodium lactate and carbonic acid.

The carbonic acid is dissociated into water and carbon dioxide, which is removed by lung ventilation.

The result is lowering of bicarbonate by 5mmol and presence of 5mmol of unmeasured anion (lactate), with no changes in sodium or chloride.

So, anion gap is increased.

In contrast, **diarrhoea** results in the loss of bicarbonate.

NaCl is reabsorbed more from kidney tubules to maintain the extracellular volume, resulting in the increase in serum chloride.

This chloride compensates for the fall in bicarbonate.

So, **diarrhea results in hyperchloremic, normal anion gap, metabolic acidosis.**

Types of Lactic Acidosis

Type A: Impaired lactic acid production with hypoxia.

Tissue hypoxia (anaerobic metabolism); Shock (anaphylactic, septic, cardiac); Lung hypoxia, Carbon monoxide poisoning, seizures

Type B: Impaired lactic acid metabolism without hypoxia.

Liver dysfunctions (toxins, alcohol); Thiamine deficiency (defective pyruvate dehydrogenase)

High Anion Gap Metabolic Acidosis (HAGMA) (Organic Acidosis)

Cause	Remarks
Renal failure	Sulfuric, phosphoric, organic anions. Decreased ammonium ion formation. Na^+/H^+ exchange results in decreased acid excretion.
Ketosis	Acetoacetate; beta hydroxy butyrate anions. Seen in diabetes mellitus starvation.
Lactic	Lactate anion accumulates acidosis when the rate of production exceeds rate of consumption.
Salicylate	Aspirin poisoning

High Anion Gap Metabolic Acidosis (HAGMA) (Organic Acidosis)

Amino Acidurias	Acidic metabolic intermediates. Accumulation due to block in normal metabolic pathway.
Organic	Organic acids (methyl acidurias malonic acid, propionic acid)
Methanol	Formate, glycolate, Oxalate ions.
Drugs	Corticosteroids, Dimercaprol, Furosemide, Nitrates, Salicylates, Thiazides

Normal Anion Gap Metabolic Acidosis (NAGMA)

When there is a loss of both anions and cations, the anion gap is normal, but acidosis may prevail.

- i. **Diarrhea**: Loss of intestinal secretions lead to acidosis. Bicarbonate, sodium and potassium are lost.
- ii. Hyperchloremic acidosis may occur in **renal tubular acidosis**, acetazolamide (carbonic anhydrase inhibitor) therapy, and ureteric transplantation into large gut (done for bladder carcinoma).

- a. Renal tubular acidosis: failure to excrete acid or reabsorb bicarbonate.
- b. Chloride is elevated since electrical neutrality has to be maintained.
- c. In ureteric transplantation, the chloride ions are reabsorbed in exchange for bicarbonate ions lost, leading to hyperchloremic acidosis.
- d. Acetazolamide therapy results in metabolic acidosis because HCO_3^- generation and H^+ secretion are affected.

Urine Anion Gap (UAG)

is useful to estimate the ammonium excretion. It is calculated as

$$\text{UAG} = \text{UNa} + \text{UK} - \text{UCl}$$

The normal value is -20 to -50 mmol/L.

In metabolic acidosis, the NH_4Cl excretion increases, and UAG becomes -75 or more.

But in RTA, ammonium excretion is defective, and UAG has positive value.

Normal Anion Gap Metabolic Acidosis (NAGMA) (Inorganic Acidosis)

Cause	Remarks
Diarrhea, intestinal fistula	Loss of bicarbonate and cations. Sodium or Potassium or both.
RTA	Defective acidification of Type I urine. urine pH is >5.5 or distal with hypokalemia.
RTA	Due to inability to reabsorb bicarbonate. Compensatory increase in chloride (hyperchloremic acidosis).

Normal Anion Gap Metabolic Acidosis (NAGMA) (Inorganic Acidosis)

Cause	Remarks
Type II or proximal RTA	urine pH is <5.5 , K normal. Due to inability to excrete hydrogen ions.
Type IV	Resistance to aldosterone, urine pH <5.5 , hyperkalemia.
Carbonic anhydrase inhibitors	Loss of bicarbonate, Na and K. Similar to proximal RTA

Normal Anion Gap Metabolic Acidosis

Cause

Remarks

Uretero-
sigmoido-
stomy

Loss of bicarbonate and reabsorption of chloride. Hyperchloremic acidosis.

Drugs

Antacids containing magnesium, Chlorpropamide, Iodide (absorbed from dressings), Lithium, Polymixin B

Causes of Renal Tubular Acidosis

Type I (Proximal RTA)

Multiple myeloma, Amyloidosis, Heavy metals; lead, mercury, Wilson's disease Galactosemia, Hyperparathyroidism Paroxysmal nocturnal hemoglobinuria Acetazolamide

Type II (Distal RTA)

Autoimmune disorders; SLE, rheumatoid Hypercalciuria, Amphotericin B, Lithium Obstructive uropathy, Marfan's syndrome

Type IV

Impaired aldosterone secretion

Decreased Anion Gap is seen in

Hypo albuminemia

Multiple myeloma (paraproteinemia)

Bromide intoxication

Hypercalcemia

Osmolal Gap

Difference between the measured plasma osmolality and the calculated osmolality,

$$2 \times [\text{Na}] + [\text{glucose}] + [\text{urea}]$$

The normal osmolal gap is <10 mOsm.

A high osmolal gap (> 25) implies the presence of unmeasured osmoles such as alcohol, methanol, ethylene glycol, etc.

Acute poisoning = a raised anion gap metabolic acidosis and an increased plasma osmolal gap. Poisoning with methanol and ethylene glycol should be considered. They are metabolized to formic acid and oxalic acids correspondingly.

Methanol will produce blindness. Ethylene glycol will lead to oxalate crystalluria and renal failure.

Compensated Metabolic Acidosis

Decrease in pH in metabolic acidosis stimulates the respiratory compensatory mechanism and produces hyperventilation (Kussmaul respiration) to eliminate carbon dioxide leading to hypocapnia (Hypocarbica).

The $p\text{CO}_2$ falls and this would attempt to restore the ratio towards 20 (partial compensation).

Compensated Metabolic Acidosis Renal compensation:

Increased excretion of acid, NH_4^+ excretion and bicarbonate reabsorption are increased.

As much as 500 mmol acid is excreted per day. The reabsorption of more bicarbonate also helps to restore the ratio to 20.

Renal compensation sets in within 2 to 4 days.

But unless the cause is also corrected, restoration of normalcy cannot occur.

Associated hyperkalemia is commonly seen due to a redistribution of K^+ and H^+ .

The intracellular K^+ comes out in exchange for H^+ moving into the cells.

Hence care should be taken while correcting acidosis which may lead to sudden hypokalemia.

Especially when treating diabetic ketoacidosis by giving glucose and insulin together.

Clinical Features of Metabolic Acidosis

The respiratory response to metabolic acidosis is to hyperventillate. So there is marked increase in respiratory rate and depth of respiration; this is called as **Kussmaul respiration**.

The acidosis is dangerous when pH is <7.2 and serum bicarbonate is <10 mmol/L.

In such conditions, there is depressed myocardial contractility.

Treatment of Metabolic Acidosis

In ketoacidosis, give intravenous fluids, insulin and potassium replacement.

Oxygen is given in patient with lactic acidosis.

In all cases, potassium abnormalities should be carefully treated.

Bicarbonate Requirement

$\text{mEq of base needed} = \text{body wt in Kg} \times 0.2 - \text{base excess in mEq/L}.$

Metabolic Alkalosis

Primary excess of bicarbonate.

Alkalosis occurs when

- a) excess base is added,
- b) base excretion is defective or
- c) acid is lost.

All these will lead to an excess of bicarbonate, so that the ratio becomes more than 20.

Metabolic Alkalosis

Primary excess of bicarbonate.

Loss of acid may result from severe vomiting or gastric aspiration leading to loss of chloride and acid. Therefore, hypochloremic alkalosis results.

Hypokalemia is closely related to metabolic alkalosis.

In alkalosis, there is an attempt to conserve hydrogen ions by kidney in exchange for K^+ . This potassium loss can lead to hypokalemia.

Subclassification of Metabolic Alkalosis

Chloride responsive conditions, urinary chloride is less than 10 mmol/L. It is seen in prolonged vomiting, naso-gastric aspiration or diuretics.

Chloride resistant condition, urine chloride is greater than 10 mmol/L; it is seen in hypertension, hyper-aldosteronism, severe potassium depletion and *Cushing's syndrome*.

Exogenous base which is often **iatrogenic**.

Metabolic Alkalosis

Type	Causes	Changes
Chloride Responsive Alkalosis; Contraction alkalosis	Prolonged vomiting, Nasogastric suction, Upper GI obstruction	Urine Chloride <10 mmol/L; Hypovolemia, increased loss of Cl, K, H ions. Increased reabsorption of Na with bicarbonate; Loss of H⁺ and K⁺. Hypokalemia leads to alkalosis due to H⁺-K⁺ exchange. Cl is reabsorbed along with Na. Hence urine chloride is low. Alkalosis responds to administration of NaCl
Loop diuretics	Blocks reabsorption of Na, K and Cl	Aldosterone secretion occurs causing Na retention and wastage of K⁺ and H⁺

Metabolic Alkalosis

Type	Causes	Changes
Chloride resistant metabolic alkalosis	Mineralocorticoid excess, Primary and secondary hyperaldosteronism, Glucocorticoid excess, Cushing's, Adrenal tumor	Urine chloride > 20 mmol/L Defective renal Cl⁻ reabsorption Associated with an underlying cause where excess mineralocorticoid activity results in Bartter's syndrome, increased sodium retention with wastage of H and K ions at the renal tubules
Exogenous base	Intravenous bicarbonate, Massive blood transfusion, Antacids, Milk alkali syndrome, Sodium Citrate overload	Excess base enters the body or potential generation of bicarbonate from metabolism of organic acids like lactate, ketoacids, citrate and salicylate

Clinical Features of Metabolic Alkalosis

The respiratory center is depressed leading to **hypoventilation**. Accumulation of CO_2 in an attempt to lower the $\text{HCO}_3^- / \text{H}_2\text{CO}_3$ ratio.

The renal mechanism conserves H^+ and **excretes more HCO_3^-** . However, complete correction of alkalosis will be effective only if potassium is administered and the cause is removed

Increased neuromuscular activity when pH is above 7.55. **Tetany** results even in the presence of normal serum calcium.

Respiratory Acidosis

Primary excess of carbonic acid. CO₂ retention as a result of hypoventilation. The ratio of bicarbonate to carbonic acid will be less than 20.

Bronchopneumonia or status asthmaticus. Overdose of sedatives or narcotics may also lead to hypercapnia.

Chronic obstructive lung disease will lead to chronic respiratory acidosis, where the fall in pH will be minimal.

Lab Findings in Respiratory Acidosis

pH

pCO₂

HCO₃⁻

Acute
respiratory
Acidosis

↓ ↓

N or

Chronic
respiratory
acidosis
(partially
compensated)

↓

Respiratory Alkalosis

Primary deficit of carbonic acid. Hyperventilation will result in washing out of CO_2 .

Bicarbonate : carbonic acid ratio is > 20 .

Hysterica, raised intracranial pressure and brain stem injury.

Early stage of salicylate poisoning causes respiratory alkalosis due to stimulation of respiratory center. But later, it ends up in metabolic acidosis due to accumulation of organic acids, lactic and keto acids. Lung diseases (pneumonia).

Respiratory Alkalosis

The $p\text{CO}_2$ is low, pH is high and bicarbonate level increases.

But bicarbonate level falls, when compensation occurs.

Clinically, **hyperventilation**, muscle cramps, tingling and **paraesthesias** are seen.

Alkaline pH will favor increased binding of calcium to proteins, resulting in a decreased ionized calcium, leading to paraesthesia.

Stepwise Approach to a Patient with Acid-base Disturbance

1. Acidemic or alkalemic
2. Primary metabolic or respiratory
3. If respiratory, whether acute or chronic
4. For metabolic acidosis, calculate anion gap
5. HAGMA or NAGMA
6. Assess whether compensatory change is within the predicted limits
7. Decide whether a simple disorder or mixed disorder
8. Check whether the values agrees with your clinical judgement

Maximum Limits of Compensation

- Metabolic acidosis, $p\text{CO}_2 = 15 \text{ mm Hg}$
- Metabolic alkalosis, $p\text{CO}_2 = 50 \text{ mm Hg}$
- Respiratory acidosis, bicarbonate = 32 mmol/L
- Respiratory alkalosis, bicarbonate = 15 mmol/L.

Normal Serum Electrolyte and Arterial Blood Gas Values

✓	pH	=	7.4
✓	Bicarbonate	=	22–26 mmol/L
✓	Chloride	=	96–106 mmol/L
✓	Potassium	=	3.5–5 mmol/L
✓	Sodium	=	136–145 mmol/L
✓	pO ₂	=	95 (85–100) mm Hg
✓	pCO ₂	=	40 (35–45) mm Hg

Respiratory Acidosis

- A. Pneumonia**
- B. Bronchitis, Asthma , COPD**
- C. Pneumothorax**
- D. Narcotics, Sedatives**
- E. Paralysis of respiratory muscles**
- F. CNS trauma**
- G. Ascites, Peritonitis**

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Metabolic Acidosis

- i. High anion gap**
 - A. Diabetic ketosis**
 - B. Lactic acidosis**
 - C. Renal failure**
- ii. Normal anion gap**
 - A. Renal tubular acidosis**
 - B. CA Inhibitors**
 - C. Diarrhea**

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- C. Pneumothorax**
- D. Narcotics, Sedatives**
- E. Paralysis of respiratory muscles**
- F. CNS trauma**
- G. Ascites, Peritonitis**

Respiratory Alkalosis

- A. High altitude**
- B. Hyperventilation**
- C. Hysteria**
- D. Febrile conditions**
- E. Septicemia**
- F. Meningitis**
- G. Congestive Cardiac Failure**

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 - B. Lactic acidosis**
 - C. Renal failure**
- ii. Normal anion gap**
 - A. Renal tubular acidosis**
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 - C. Diarrhea**

Respiratory Acidosis

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- ii. Normal anion gap**
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 - C. Diarrhea**

Metabolic Alkalosis

- A. Severe vomiting**
- B. Cushing Syndrome**
- C. Milk alkali syndrome**
- D. Diuretic therapy (K loss)**